An Arithmetic Stratification of Polynomial Continued Fractions: Program Statement of the SIARC Series

(v2.1: Closure Cascade Amendment)

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Abstract

This paper is a *program statement*, not a results paper. It is the v2.0 revision of the SIARC umbrella program statement, refactoring the v1 (April 2026) framing around a cross-degree *invariant triple* $\big(\Delta_d(b),\ \|\Delta\|_{\mathrm{Pet}}(\tau_b),\ \xi_0(b)\big)$ that emerged from the SIARC stack between v1 and v2. The triple combines the polynomial discriminant of the denominator polynomial b (the discriminant axis), the Petersson-normalised modular discriminant evaluated at the period τ_b of the curve $y^2 = b(x)$ (the modular-discriminant axis), and the Newton-polygon Borel radius $\xi_0(b) = d/\beta_d^{1/d}$ (the Borel-radius axis). v2.0 integrates the four sister Zenodo records published since v1 — PCF-1 v1.3, PCF-2 v1.3, Channel Theory v1.2, and the standalone j-invariant note — into a single cross-degree organisational scheme, retains the $\mathrm{PSL}_2(\mathbb{Z})$ four-tier hierarchy at degree pair $(2, 1)$ as the d=2 specialisation, and refreshes the Open Problems list with the post-T2 entries (op:j-zero-amplitude-h6, op:j-1728-wedge, op:shallow-j-effect-d4, op:b4-degree-d, op:median-resurgence-extension). *No proofs and no new theorems are reproduced here*; every nontrivial claim is stated as a conjecture or attached to a companion paper by Zenodo DOI. The purpose of this document is to provide a stable, citable scope-anchor (*[SIARC v2.1]*, superseding *[SIARC v2.0]*) for the program and to fix unified notation for the post-March 2026 stack. The v2.1 revision (May 2026) appends a Closure Cascade and Bridge Provenance section (§8) recording three milestone closures landed on the SIARC relay bridge since v2.0: the M4 V0 closure (deg-$a = 0$ row mechanism, ratified 2026-05-08), the M6.CC R1 closure ($\mathrm{V}_{\mathrm{quad}}$ on the $D_6 \to D_7$ degeneration boundary, via the Q4 v2 cascade), and the Route F slot 115 numerical re-derivation (image $(\alpha, \beta, \gamma, \delta) = (0, 0, 1/9, 0)$ at dps $= 300$ with s_1 fixed-point at $\alpha_1 = 0$). *All mathematical content, conjectures, open problems, and the cross-degree triple of §3 are unchanged from v2.0*; the v2.1 amendment is a status / provenance addendum, not a content revision.

Contents

1 Introduction and Motivation

A *polynomial continued fraction* (PCF) is a formal continued fraction

$$K(x) \;=\; b_0(x) + \cfrac{a_1(x)}{b_1(x) + \cfrac{a_2(x)}{b_2(x) + \cfrac{a_3(x)}{b_3(x) + \ddots}}} \tag{1}$$

in which the partial numerators $a_n(x)$ and partial denominators $b_n(x)$ are fixed polynomials in x with integer (or rational) coefficients, evaluated at a fixed integer $x \in \mathbb{Z}_{\geq 1}$. The classical Ramanujan and Rogers–Ramanujan continued fractions, the Stern–Brocot expansions of quadratic irrationals, and the Apéry-style recurrences for $\zeta(2)$ and $\zeta(3)$ are all PCFs in this sense. A central problem, going back to Stieltjes and revisited in the modern era by Lagarias, Borwein, and others, is to determine the *arithmetic nature* of the limit of (1): when is it rational, when is it algebraic, when is it transcendental, and when does it fail to converge to a number that admits any such classification?

The SIARC program proposes that this arithmetic question is not homogeneous. PCFs admit a natural *stratification* indexed by (i) the uniformizing Fuchsian group of the associated monodromy representation, and (ii) an obstruction tier that records how far the limit lies from being algebraic over $\mathbb{Q}$. This stratification organizes otherwise unrelated empirical phenomena — the appearance of $-2/9$ exponents in degree-$(2,1)$ families, the Painlevé connection in degree-$(3,1)$ families, the failure of the Mahler criterion at specific lattice points — into a single arithmetic picture.

Goal of the program. The SIARC program aims to define, for each Fuchsian group Γ that arises as a uniformizer of a PCF family, a four-tier arithmetic stratification (§2), to verify the stratification on the $\mathrm{PSL}_2(\mathbb{Z})$ case completely (companion paper #14), and to extend it conjecturally to the remaining classical groups via the Trans-Stratum Conjecture (§5) and the open problems of §6.

What changed between v1 and v2.0. v1 (Zenodo concept DOI 10.5281/zenodo.19885550) organised PCFs by (Fuchsian group, tier), anchored in $\mathrm{PSL}_2(\mathbb{Z})$ and the degree pair $(2,1)$. Between v1 and v2.0 the SIARC stack produced four companion records that demand a cross-degree organisational principle: PCF-1 v1.3 (sharp WKB exponents at $d = 2$ stratified by

$\mathrm{sgn}(\Delta_2)$), PCF-2 v1.3 (a cubic-modular framing at $d = 3$), Channel Theory v1.2 (a Borel-resurgence functor on the $\mathrm{V}_{\mathrm{quad}}$ stratum), and a standalone j-invariant note (the equianharmonic cubic cell). v2.0 reorganises the umbrella around the *invariant triple*

$$\mathrm{PCF}(1, b) \;\longmapsto\; \big(\Delta_d(b),\ \|\Delta\|_{\mathrm{Pet}}(\tau_b),\ \xi_0(b)\big), \tag{2}$$

introduced in §3. The two pictures are *compatible*: at $d = 2$ the triple-invariant framing recovers the $\mathrm{PSL}_2(\mathbb{Z})$ four-tier hierarchy of v1 as the Δ_2-axis split with trivial $\|\Delta\|_{\mathrm{Pet}}$-axis, and the Trans-Stratum two-class conjecture of §5 reappears as the Class A / Class B dichotomy at $d = 2$. At $d \geq 3$ the triple is genuinely new information; the four-tier hierarchy extends *conjecturally*, with empirical support from PCF-2 v1.3.

What changed between v2.0 and v2.1. v2.1 (May 2026) is a status / provenance addendum, not a content revision. No definition, conjecture, open problem, or theorem cited in v2.0 is altered. The v2.1 amendment appends a Closure Cascade and Bridge Provenance section (§8) bundling three milestone closures landed on the SIARC relay bridge (github.com/papanokechi/siarc-relay-bridge) since v2.0: M4 V0 (deg-$a = 0$ row mechanism, ratified 2026-05-08; bridge SHA `5f9db69`); M6.CC R1 ($\mathrm{V}_{\mathrm{quad}}$ on the $D_6 \to D_7$ degeneration boundary via the Q4 v2 cascade; bridge SHAs `10b5cf6` + `8a22b11`); and the Route F slot 115 numerical re-derivation of the $\mathrm{V}_{\mathrm{quad}}$ image $(\alpha, \beta, \gamma, \delta) = (0, 0, 1/9, 0)$ at dps $= 300$ with the s_1 fixed-point at $\alpha_1 = 0$ under the surviving $\widetilde{W}_a(A_1)$ Cremona action (bridge SHA `8ebd1eb`). All three closures are inherited from companion papers and bridge artefacts; no numerical claim originates in this v2.1 document.

2 The Four-Tier Hierarchy

We summarize the tier definitions. Full proofs of the classification theorem are deferred to the $\mathrm{PSL}_2(\mathbb{Z})$ paper; see Remark 6.

Definition 1 (Tier T0: Algebraic)**.** A PCF family $\{K(x)\}_{x\in S}$ lies in tier T0 over an arithmetic set $S \subseteq \mathbb{Z}_{\geq 1}$ if $K(x)$ is algebraic over $\mathbb{Q}$ for every $x \in S$, and the algebraic degree is uniformly bounded as x varies in S.

Definition 2 (Tier T1: Transcendental)**.** A PCF family lies in tier T1 if $K(x)$ is transcendental for every $x \in S$, *and* a Mahler-type or Nesterenko-type criterion certifies the transcendence uniformly in x.

Definition 3 (Tier T2: Trans-stratum)**.** A PCF family lies in tier T2 if its irrationality measure $\mu(K(x))$ satisfies an asymptotic

$$\mu(K(x)) \;=\; 2 + \frac{c}{\log x} + o\!\left(\frac{1}{\log x}\right), \qquad x \to \infty,$$

for an explicit rational constant c (the *trans-stratum exponent*) that depends only on the polynomial degrees $(\deg a, \deg b)$ and the Fuchsian group Γ. The trans-stratum is therefore not a transcendence statement, but a statement about the rate at which transcendence is approached.

Definition 4 (Tier T3: Barrier)**.** A PCF family lies in tier T3 if the Fuchsian uniformization itself degenerates: either Γ fails to be of finite covolume, or the associated L-function fails to admit analytic continuation to the required half-plane. T3 families are inaccessible to the methods of T0–T2 and serve as the negative boundary of the program.

Remark 5 (Translation to the obstruction hierarchy of #14). The companion paper #14 introduces a parallel four-tier *obstruction* hierarchy $O_0, \ldots, O_3$ measuring the distance from the modular sector $X(1)$. The two hierarchies do not coincide. We have the inclusions $O_0 \subseteq T_0 \cup T_2$,

$O_1 \subseteq T_2$, $O_2 \subseteq T_2$, $O_3 \subseteq T_3$, but in general neither family of tiers refines the other. We use $T_0, \ldots, T_3$ exclusively for the arithmetic hierarchy of this paper, and $O_0, \ldots, O_3$ for the obstruction hierarchy of #14.

Remark 6 (Source of the classification). A full classification theorem stating that every $\mathrm{PSL}_2(\mathbb{Z})$-uniformized PCF family lies in exactly one of T0, T1, T2, T3 — and that the tier is computable from $(\deg a, \deg b)$ and the leading coefficients — is the main theorem of companion paper #14. We do not reproduce either the statement in full generality or the proof here. See the companion paper table in §7.

Forward reference to the cross-degree framing. The four tiers above are stated for $\mathrm{PSL}_2(\mathbb{Z})$-uniformized families at degree pair $(2, 1)$. Their behaviour at degree $d \geq 3$ is the subject of the cross-degree framing of §3: the discriminant axis Δ_d refines the T0/T1 boundary at every d (PCF-1 v1.3 Theorem 5; [1]); the modular-discriminant axis $\|\Delta\|_{\mathrm{Pet}}$ refines the T2 stratum at $d = 3$ (PCF-2 v1.3 Conjectures B5/B6; [2]); the Borel-radius axis ξ_0 refines the T3 boundary via Newton-polygon resurgence (Channel Theory v1.2; [3]). The v1 four-tier hierarchy is the d=2, Δ_2-only specialisation of the cross-degree picture.

3 Cross-Degree Framing: the Invariant Triple

This section is new in v2.0. It records the cross-degree organisational principle that emerged from the SIARC stack between April and May 2026. *No new theorems are stated here*; every quantitative claim is either empirical (with citation to the responsible companion paper's session deliverables) or conjectural.

3.1 Definitions

Throughout this section, fix a PCF family of the form $\mathrm{PCF}(1, b) = 1/(b_1 + 1/(b_2 + \cdots))$ with $b_n = b(n)$ for a fixed polynomial $b \in \mathbb{Q}[x]$ of degree d. Write $b(x) = \beta_d x^d + \beta_{d-1}x^{d-1} + \cdots + \beta_0$.

Definition 7 (Discriminant axis)**.** The *discriminant* of b is $\Delta_d(b) := \mathrm{disc}(b)$, the ordinary polynomial discriminant of the squarefree-part-stripped denominator polynomial. By convention $\Delta_d(b) \in \mathbb{Q}^\times$ when b is squarefree, and the sign $\mathrm{sgn}(\Delta_d) \in \{\pm 1\}$ is the discrete invariant on which PCF-1 v1.3 stratifies the sharp WKB exponent at $d = 2$.

Definition 8 (Modular-discriminant axis)**.** For $d = 3$ with b irreducible non-singular, the curve $E_b : y^2 = b(x)$ is an elliptic curve over $\mathbb{Q}$; let $\tau_b \in \mathfrak{H}/\mathrm{SL}_2(\mathbb{Z})$ be its period in the standard fundamental domain. The *Petersson-normalised modular discriminant* at τ_b is

$$\|\Delta\|_{\mathrm{Pet}}(\tau_b) \ := \ |\Delta(\tau_b)| \cdot (\mathrm{Im}\, \tau_b)^6, \tag{3}$$

where $\Delta(\tau) = (2\pi)^{12}\eta(\tau)^{24}$ is the modular discriminant cusp form of weight 12. For general $d \geq 3$ the analogous construction uses the period of the hyperelliptic curve $y^2 = b(x)$ on the corresponding period domain; we restrict the quantitative claims of this section to $d = 3$.

Definition 9 (Borel-radius axis)**.** The *Borel radius* of the PCF asymptotic series associated to $\mathrm{PCF}(1, b)$ is

$$\xi_0(b) \ := \ \frac{d}{\beta_d^{1/d}}, \tag{4}$$

the Newton-polygon root extracted from the dominant balance of the characteristic equation of the three-term recurrence at infinity ([2], Q1-B Proposition; [3], $\mathrm{V}_{\mathrm{quad}}$ recovery).

The map

$$\Phi : \mathrm{PCF}(1, b) \ \longmapsto \ \big(\Delta_d(b),\ \|\Delta\|_{\mathrm{Pet}}(\tau_b),\ \xi_0(b)\big) \tag{5}$$

is the *bridge functor* of v2.0; the rest of this section reviews what each axis carries.

3.2 The discriminant axis Δ_d

At $d = 2$, PCF-1 v1.3 Theorem 5 establishes the sharp WKB exponent identity for PCF$(1, b)$ with $\deg b = 2$, with the exponent split by $\mathrm{sgn}(\Delta_2)$: $A = 4$ for $\Delta_2 > 0$ and $A = 3$ for $\Delta_2 < 0$ (six $\Delta_2 < 0$ families verified at ≥ 200 digits; see PCF-1 v1.3 [1]). At $d \geq 3$, PCF-2 v1.3 [2] verifies the unsplit *sharp* Conjecture B4

$$\log|\delta_n| \;\sim\; -A\,n\log n + \alpha\,n - \beta\log n + \gamma, \qquad A = 2d \tag{6}$$

on the joint $d \in \{3, 4\}$ catalogue (110/110 families at the relevant tail-window precision). The strong-form upgrade op:b4-degree-d (§6, prob:b4-allD) extends $A = 2d$ to all d via the Birkhoff–Trjitzinsky asymptotic theory of irregular linear difference equations [8, 9]; this remains open and is the gating move toward the *SIARC Master Conjecture v0* (MASTER-V0, §7).

3.3 The modular-discriminant axis $\|\Delta\|_{\mathrm{Pet}}$

The defining empirical content of v2.0 is the post-T2 finding ([2], Session T2; bridge mirror at PCF2-SESSION-T2) that, on the deep R1.3 cubic catalogue ($n = 50$ cubic families, dps $= 4000$, $N_{\max} = 480$), the Petersson-normalised modular discriminant correlates with the residual $\delta = A_{\mathrm{fit}} - 6$ with Spearman

$$\rho\big(\log\|\Delta\|_{\mathrm{Pet}}(\tau_b),\ \delta_{\mathrm{deep}}\big) \;=\; +0.638, \qquad p_{\mathrm{Bonf}}(K{=}14) \;=\; 8.6\times10^{-6},$$

beating the bare-log $|j|$ baseline $\rho = -0.568$, $p_{\mathrm{Bonf}} = 2.34\times10^{-4}$ by a factor of ~ 30 in Bonferroni p. Bare $\log|E_4(\tau_b)|$ alone *underperforms* $\log|j|$ ($\rho = -0.459$, $p_{\mathrm{Bonf}} = 1.12\times10^{-2}$). This pattern is exactly the prediction of Fleet-A H2: the local mechanism at the equianharmonic ($j = 0$) cell is the simple zero $E_4(\rho) = 0$, so $\log|E_4|$ is locally pinned while $\log\|\Delta\|_{\mathrm{Pet}} = 6\log\mathrm{Im}\,\tau + \log|\Delta|$ remains well-conditioned globally via the modular identity $1728\,\Delta = E_4^3 - E_6^2$. Compactly:

Conjecture 10 (Cubic-modular split, $d = 3$)**.** *For* PCF$(1, b)$ *with* $b \in \mathbb{Q}[x]$ *irreducible non-singular cubic:*

- ***B5*** ***($d = 3$ form):*** $j(E_b) = 0$ *implies* $A_{\mathrm{PCF}}(b) = 2d = 6$ *exactly, with subleading amplitude conjecturally in the Chowla–Selberg* $\Gamma(1/3)$ *basis [10].*
- ***B6*** ***($d = 3$ form):*** $\rho_{\mathrm{Spearman}}\big(\log\|\Delta\|_{\mathrm{Pet}}(\tau_b), A_{\mathrm{fit}}(b) - 6\big) > 0$ *on the* $n = 50$ *cubic catalogue, with* $p_{\mathrm{Bonf}} \leq 10^{-5}$ *at* $K = 14$.

The d=3 restriction is the verdict of Session R1.3 (PCF2-SESSION-R1.3, CASE B with C-caveat): the deep-WKB null at $d = 4$ (family 32 deep $\delta = -4.55\times10^{-4}$ vs family 1 baseline deep $\delta = -4.60\times10^{-4}$, $|\mathrm{diff}|$ at 0.24 pooled stderr units) shows that the sharp/deep-WKB form of B5/B6 is $d = 3$-specific. A residual shallow-N $j = 0$-cell shift at $d = 4$ exists but does not survive to deep-N; this is logged as op:shallow-j-effect-d4 (§6). The Phase D finite-N tail-window artefact at the four $j = 0$ cubic families ($|\delta| \sim 1.5\times10^{-5}$, shrinking $\geq 50\times$ from the R1.1 $N \leq 67$ window to the T2-D $N \leq 480$ window) is the content of op:j-zero-amplitude-h6: a closed-form Chowla–Selberg amplitude is conjectured but requires a redesigned five-parameter WKB ansatz to extract.

3.4 The Borel-radius axis ξ_0

The identity $\xi_0(b) = d/\beta_d^{1/d}$ is the master Newton-polygon characteristic-root identity (PCF-2 v1.3 Q1-B Proposition; [2]). At $d = 4$ it is verified to spread 0 across 8 Q1 representatives; at $d = 3$ it is the same identity used in PCF-1 v1.3's exponent split. The Channel Theory companion ([3], $\mathrm{V_{quad}}$ recovery) extends this to a functor on the Borel plane: the $\mathrm{V_{quad}}$ CC-channel branch-point structure is recovered to 200 digits via Newton-polygon + Birkhoff–Trjitzinsky resurgence, and Fleet-A H4 confirms that median resurgence delivers $\geq$ 30-digit accuracy on the V_{quad} stratum. The extension of median resurgence beyond V_{quad} to the cubic CC channels is open (prob:median-resurgence-extension).

3.5 The $E/P_k/B$ cell decomposition

The triple $(\Delta_d, \|\Delta\|_{\text{Pet}}, \xi_0)$ partitions PCF families into three structural cells:

- **Cell E (generic / elliptic):** $\Delta_d \neq 0$, $\|\Delta\|_{\text{Pet}}(\tau_b)$ generic, ξ_0 on the principal Newton-polygon branch. The bulk of the cubic catalogue lives here.
- **Cell P_k (Painlevé reduction at level k):** ξ_0 degenerates to a Painlevé-VI special-parameter substratum at level k, in the sense of P08 [7]. The $D = 2$ Painlevé reduction at degree pair $(2, 1)$ is the historical entry point; higher k is open (prob:Pk-cells-structure; Fleet-A H3 status: D=2_REDUCTION_AMBIGUOUS).
- **Cell B (Borel-channel-anomalous):** the V_{quad} stratum and its conjectural cubic-CC-channel extensions; characterised by the channel functor χ of [3].

The bridge functor Φ of (5) factors through the channel functor χ in a way conjectured to be compatible with the cell decomposition (op:chi-Phi-compatibility; prob:Pk-cells-structure). A worked example pairing χ and Φ on a single E-cell representative is deferred to D2-NOTE (§7); v2.0 only fixes the notation.

4 Logical Structure of the Program

4.1 Companion papers and dependency graph

Figure 1 shows the dependency structure of the program after the post-March 2026 stack. The $\mathrm{PSL}_2(\mathbb{Z})$ classification (#14) and the Trans-Stratum paper (T2B) remain at the centre; PCF-1 / PCF-2 / Channel Theory enter as the cross-degree extension spine; the standalone j-invariant note (jSN) is the polished exposition of the equianharmonic cubic cell of §3.3.

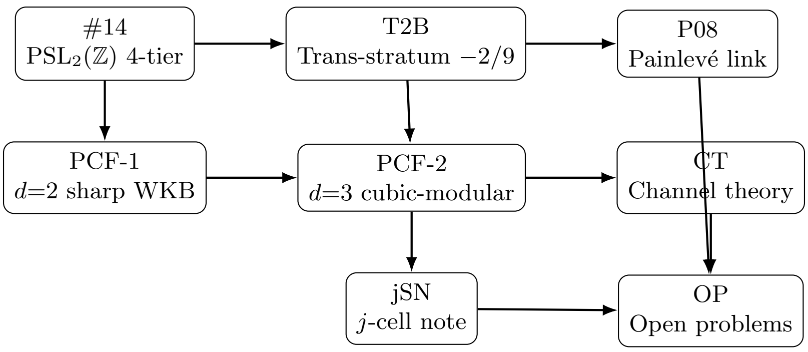

Figure 1: Dependency graph of the SIARC companion papers (v2.0). Arrows denote "logical prerequisite"; the umbrella paper (this document, v2.0) sits above all nodes and provides the unified notation and the cross-degree invariant triple of §3.

4.2 Unified notation

Across the series we adopt the following conventions. Companion papers that predate this umbrella may use slightly different notation locally; the table below is the canonical reference and absorbs the v2.0 additions (the invariant triple, the channel and bridge functors, the cell decomposition, and the residual notation).

Symbol	Meaning	Convention
x	PCF variable	integer, $x \geq 1$
$a_n(x), b_n(x)$	partial num./denom. polynomials	$\mathbb{Z}[x]$
(d_a, d_b)	degree pair	$d_a = \deg a_n$, $d_b = \deg b_n$
d	denominator polynomial degree	$d = \deg b$ (cross-degree label)
Γ	uniformizing Fuchsian group	subgroup of $\mathrm{PSL}_2(\mathbb{Z})$ unless stated
$K(x)$	limit of the PCF at x	in $\mathbb{R} \cup \{\infty\}$
$\mu(\alpha)$	irrationality measure of α	$\mu \geq 2$ for irrationals
$c_{\Gamma,(d_a,d_b)}$	trans-stratum exponent	rational, depends on Γ and degrees
T0–T3	arithmetic tiers	as in §2
$\Delta_d(b)$	polynomial discriminant axis	$\mathrm{disc}(b)$, §3.2
τ_b	period of $E_b : y^2 = b(x)$	in $\mathfrak{H}/\mathrm{SL}_2(\mathbb{Z})$
$\|\Delta\|_{\mathrm{Pet}}(\tau_b)$	Petersson modular-discriminant axis	$\lvert\Delta(\tau_b)\rvert(\operatorname{Im}\tau_b)^6$
$\xi_0(b)$	Borel-radius axis	$d/\beta_d^{1/d}$, §3.4
$E,\ P_k,\ B$	cell decomposition	§3.5
χ	channel functor	[3]
Φ	bridge functor (umbrella v2.0)	(5)
δ_n	PCF residual (PCF-2 notation)	$\delta_n = \log\lvert p_n - K(x)q_n\rvert$
$A_{\mathrm{fit}}, A_{\mathrm{PCF}}$	fitted and conjectured WKB exponent	$A_{\mathrm{PCF}} = 2d$ (Conj. B4)

Degree convention. Throughout the series, "degree-(d_a, d_b)" refers to the *generic* polynomial degrees of a_n and b_n as functions of the index n in the underlying recurrence, not as functions of x. A degree-$(2, 1)$ family is therefore one in which the numerator polynomial grows quadratically in n and the denominator grows linearly in n. The cross-degree label $d = \deg b$ of §3 matches d_b at $\mathrm{PCF}(1, b)$ where $\deg a = 0$.

Coefficient ordering. PCF family coefficients $[a_2, a_1, a_0, b_1, b_0]$ and $[\beta_d, \ldots, \beta_0]$ are stored *leading-coefficient-first* throughout the SIARC stack (`f1_base_computation.py` convention). This matches the v2.0 notation: $\xi_0(b) = d/\beta_d^{1/d}$ uses β_d as the leading coefficient of b.

5 The Trans-Stratum Conjecture

The single most distinctive empirical phenomenon of the program is the persistent appearance of the rational exponent $-2/9$ in the trans-stratum of degree-$(2, 1)$ families uniformized by $\mathrm{PSL}_2(\mathbb{Z})$.

Conjecture 11 (Trans-stratum two-class conjecture)**.** *Let $\{K(x)\}$ be a PCF family of generic degree $(2, 1)$ uniformized by $\mathrm{PSL}_2(\mathbb{Z})$, satisfying the regularity conditions of companion paper T2B. Then $K(x)$ lies in tier T2 and falls into exactly one of two classes, distinguished by the structural ratio $\rho := a_2/b_1^2$:*

- *Class A ($\rho = -2/9$): trans-stratum exponent $c_{\mathrm{PSL}_2(\mathbb{Z}),(2,1),A} = -2/9$ (T2B Theorem 2, conditional on $S_{12} \neq 0$);*
- *Class B ($\rho = +1/4$): limits lie in $\mathbb{Q}(\pi)$ (T2B Theorem 3); the Pure-regime subclass satisfies the stronger $\mathbb{Q} \cdot \pi^{-1}$ (T2B Cor. classB-pi).*

Equivalently (T2B Completeness Conjecture): the degree-$(2, 1)$ Trans-stratum is exactly $\mathcal{T}_A \sqcup \mathcal{T}_B$, with no third class.

Remark 12 (Pre-revision form). The version of this conjecture in UMB v1 (Zenodo 10.5281/zenodo.19885550) stated only the Class A clause. The current form incorporates the Class B census of companion paper T2B (16 canonical families, height ≤ 100). No previously asserted Class A claim is revoked; the change is purely additive.

Remark 13 (Empirical basis). A computational census reported in companion paper T2B (Zenodo concept DOI 10.5281/zenodo.19783311) covers approximately 150,000 degree-$(2,1)$ families generated under the regularity conditions of that paper. Across the entire census, no counterexample to Conjecture 11 has been observed; the empirical fit to the predicted $-2/9$ slope is consistent with the reported numerical precision. We do not reproduce the empirical methodology, the regularity conditions, or any proof attempt in this document; all such material lives in T2B.

Remark 14 (Why $-2/9$?). The value $-2/9$ arises in companion paper T2B via the indicial exponents $\{1/3, 2/3\}$ of the associated three-term recurrence at infinity: the ratio a_2/b_1^2 is determined by requiring the indicial roots to sum to 1 and multiply to $2/9$, giving $1/3+2/3=1$ and $(1/3)(2/3)=2/9$. A congruence-subgroup interpretation of this denominator is speculative and not established; a rigorous derivation from the monodromy of the uniformizing group remains the central open problem.

Remark 15 (Projective verification of Conjecture 11, Class A). A targeted projective-coordinate sweep ($H=8$, $b_1 \in \{3, 6, \ldots, 30\}$ subject to $3 \mid b_1$, 48,552 primitive 5-tuples $(a_2, a_1, a_0, b_1, b_0)$ satisfying $9a_2+2b_1^2=0$, $\gcd=1$, $b_1>0$; 43,915 distinct-L families; stratified random sample of 400 groups, 40 per b_1 bucket) confirms the Class A clause of Conjecture 11 empirically: 387/400 sampled families are Trans, with 0 Log, 0 Alg, 13 rational, and 0 phantom hits under PSLQ at `dps`$=400$, tol $=10^{-50}$, basis $\{1, \pi, \pi^2, \log 2, \zeta(2), \zeta(3), \sqrt{2}, \sqrt{3}, \sqrt{5}, e, \gamma, \log 3, \pi\log 2, \pi^2, \pi^3\}$, with mandatory rejection of any relation whose L-coefficient vanishes. The smallest witness is $(a_2, a_1, a_0, b_1, b_0) = (-2, 0, 1, 3, -1)$ with $L \approx 0.26057$. An earlier affine-coordinate enumeration at fixed height $H=5$ ($b_1 \in \{9, 12, 15, 30\}$) found 0 Trans families; the discrepancy is a coordinate artefact resolved by passing to projective normalization with $\gcd=1$ and $b_1>0$, which admits primitive small-coefficient witnesses (in particular $b_1=3$ with $|a_1|, |a_0|, |b_0| \leq 1$) that are absent from any fixed-height affine slice. Every $3 \mid b_1$ bucket in the sample contains at least 32 Trans families. Full numerics, classifier source, and sample data are recorded in `sessions/2026-04-30/UMB-RES-EXTEND-PROJ/` of the SIARC relay bridge repository.

Cross-degree extension. The Trans-Stratum two-class conjecture above is the $d=2$ specialisation of the cross-degree framing of §3. Class A (rationals, $\rho=-2/9$, Birkhoff–Trjitzinsky resolution) and Class B (rationals in $\mathbb{Q}(\pi)$, $\rho=+1/4$, Gauss-continued-fraction / Ramanujan resolution) are recovered as the two branches of the Δ_2-axis split: $\Delta_2<0$ aligns with Class A and the BT-resolved tier; $\Delta_2>0$ aligns with Class B and the GCF/Ramanujan-resolved tier (PCF-1 v1.3 [1]). At $d \geq 3$, the analogous split is the Δ_d-axis component of the bridge functor Φ of (5); the $\|\Delta\|_{\mathrm{Pet}}$-axis refinement is genuinely new at $d=3$ (Conjecture 10). The v2.0 framing therefore *subsumes* the (2,1)-only Trans-Stratum Conjecture without retracting any of its content.

6 Open Problems

We list the principal open problems of the program in rough order of expected tractability. Each is stated informally; precise formulations appear in the indicated companion papers. Problems carried over from v1 retain their numbering; v2.0 additions follow.

Open Problem 1 (Degree-$(3,1)$ classification)**.** *Establish the four-tier classification for* $\mathrm{PSL}_2(\mathbb{Z})$*-uniformized PCF families of generic degree* $(3,1)$*. The asymmetry between numerator and denominator degrees breaks the symmetric treatment of #14.* Status (v2.0): *the structural-correlation half of the problem is substantially answered at* $d=3$ *by PCF-2 v1.3 [2] and the* *cubic-modular framing of §3.3 (R1.1 / T2 verdicts); the rigorous-classification half remains open. Cross-references:* op:b4-degree-d*,* prob:b4-allD*.*

Open Problem 2 (Trans-stratum at $d = 2$ for non-$\mathrm{PSL}_2(\mathbb{Z})$)**.** *Determine the trans-stratum exponent $c_{\Gamma,(2,1)}$ for $\Gamma = \Gamma_0(2)$ and for the Hecke triangle groups G_q with $q \geq 4$. Is the value $-2/9$ specific to $\mathrm{PSL}_2(\mathbb{Z})$, or does it recur with a Γ-dependent denominator?* Status (v2.0): *unchanged; the $\Gamma_0(2)$ sweep (`sessions/2026-04-30/UMB-GAMMA0-2-SWEEP/`) supplies a preliminary census but no exponent identification.*

Open Problem 3 (Painlevé connection)**.** *Make precise the conjectural correspondence between trans-stratum PCF families and solutions of Painlevé VI with monodromy in Γ. A pointer-level discussion is given in companion paper P08; a full correspondence theorem is open.*

Open Problem 4 (Effective T0/T1 boundary)**.** *Provide an algorithm that, given the polynomials $a_n(x), b_n(x)$ in exact arithmetic, decides in finite time whether the family lies in T0 or T1. #14 establishes the boundary as a set; effectivity is open.*

Open Problem 5 (T3 nonemptiness)**.** *Exhibit an explicit PCF family that provably lies in T3 (not merely fails to lie in T0–T2 by current methods). Equivalently, exhibit a PCF whose monodromy group has infinite covolume in a controlled way.*

Open Problem 6 (Mahler–Nesterenko gap)**.** *For T1 families, identify the precise hypothesis that fails when the Mahler criterion does not apply and the Nesterenko criterion does. The empirical gap is small; a structural explanation is missing.*

Open Problem 7 (Census completeness)**.** *Extend the empirical census underlying Conjecture 11 beyond degree-$(2, 1)$. A negative result — a single counterexample in any degree — would falsify the Trans-Stratum Conjecture as stated and force a refinement.*

New problems introduced by v2.0.

Open Problem 8 (Conjecture B4 at all d via Birkhoff–Trjitzinsky)**.** *Establish $A_{\mathrm{PCF}}(b) = 2d$ for all d and all irreducible non-singular $b \in \mathbb{Q}[x]$ with* $\deg b = d$*, in the sharp form of* (6)*. PCF-1 v1.3 verifies $d = 2$; PCF-2 v1.3 verifies $d \in \{3, 4\}$ on* 110/110 *families. The strong-form upgrade follows the Birkhoff–Trjitzinsky asymptotic theory of irregular linear difference equations [8, 9]; this is the gating move toward the SIARC Master Conjecture v0. Tag:* op:b4-degree-d*.*

Open Problem 9 (Modular-discriminant stratification at $d = 3$)**.** *Make precise the conjectural claim that $\|\Delta\|_{\mathrm{Pet}}(\tau_b)$ stratifies the $d = 3$ cubic catalogue beyond the bare j-invariant. Identify the modular-form weight at which the stratification becomes sharp (in particular: is the weight-*12 *Petersson discriminant the asymptotic optimum, or does a higher-weight cusp form refine it further?). Cross-references: [2] Session T2, Conjecture 10.*

Open Problem 10 (Closed-form $j = 0$ amplitude (Chowla–Selberg))**.** *Closed-form expression for the cubic $A = 6$ subleading amplitude at the equianharmonic CM cell ($j = 0$, $\tau = \rho = e^{2\pi i/3}$), conjectured to involve a Chowla–Selberg $\Gamma(1/3)$ basis element [10]. Recommended protocol: redesigned five-parameter WKB ansatz at* $\mathrm{dps} \geq 8000$, $N_{\max} \geq 1200$*, with explicit Chowla–Selberg target basis. Tag:* op:j-zero-amplitude-h6 *(relay* T2.5d*).*

Open Problem 11 ($j = 1728$ wedge)**.** *Order-2 elliptic-point CM-locus probe at $d = 3$, $j =$* 1728*: quantify the residual $\|\Delta\|_{\mathrm{Pet}}$-axis signal at the $E_6(\tau_b) = 0$ wedge after the E_4-zero ($j = 0$) cell is removed. Tag:* op:j-1728-wedge *(relay* T2.5b*).*

Open Problem 12 (Shallow-N $j = 0$ effect at $d = 4$)**.** *The shallow-N $j = 0$ cell shift at $d = 4$ is highly significant ($p = 1.1 \times 10^{-6}$ stratified) but dissolves at deep N. Identify the sub-leading WKB term ($\beta \log N$ or γ) that carries this shallow signal, and decide whether it lifts to a deep-N rule under any precision/window enlargement. Tag:* op:shallow-j-effect-d4*.*

Open Problem 13 (Painlevé cells P_k as substrata of $\mathcal{M}_{1,1}$). *Characterise the Painlevé-reduction cells P_k of §3.5 as substrata of the moduli space $\mathcal{M}_{1,1}$ (or its degree-d generalisation). Fleet-A H3 status:* D=2_REDUCTION_AMBIGUOUS *(the P_2 cell at $d = 2$ admits more than one Painlevé parametrisation under the current criteria).*

Open Problem 14 (Median-resurgence extension beyond $\mathrm{V}_{\mathrm{quad}}$). *Extend the median-resurgence framework of Channel Theory v1.2 [3] from the V_{quad} stratum to the cubic CC channels. Fleet-A H4 confirmed ≥ 30-digit recovery on V_{quad}; the cubic-CC analogue is open. Tag:* op:median-resurgence-extension.

Open Problem 15 (Compatibility of χ and Φ). *Establish that the bridge functor Φ of* (5) *factors through the channel functor χ of [3] in a way compatible with the cell decomposition $E/P_k/B$ of §3.5. Tag:* op:chi-Phi-compatibility.

Problems closed since v1. op:finer-cubic-split (resolved by PCF-2 v1.3 cubic-modular framing; closed in [2]). The resolution is Conjecture 10 above.

7 Companion Papers and Status

The table below records the current state of the SIARC series at the time of this v2.0 umbrella draft. DOIs marked "–" are not yet issued; "TBA" is pending. Statuses are intended to be updated in revisions of this document.

ID	Title (working)	Venue / status	Zenodo concept / version DOI
Published (Zenodo, May 2026)			
PCF-1	Polynomial continued fractions at degree $(d, 1)$: sharp WKB exponents and an arithmetic stratification	Zenodo v1.3 (2026-05-01); Compositio CM 10573 in review	10.5281/zenodo.19937196 (v1.3) / concept 19931635
PCF-2	Polynomial continued fractions at degree three: a cubic-modular framing	Zenodo v1.3 (2026-05-02); program statement, 22pp	10.5281/zenodo.19963298 (v1.3) / concept 19936297
CT	Channel Theory: $\mathrm{V}_{\mathrm{quad}}$ resurgence and the CC-pipeline	Zenodo v1.2 (2026-05-01)	10.5281/zenodo.19951331 (v1.2) / concept 19941678
T2B	Two arithmetic classes of degree-$(2, 1)$ Trans-stratum continued fractions: a Birkhoff–Trjitzinsky / Gauss-continued-fraction dichotomy	Zenodo v3.0 (2026-04-30)	10.5281/zenodo.19915689 (v3.0) / concept 19783311
UMB	*This document* (umbrella program statement, v2.1)	Zenodo v2.1 (this version)	concept 10.5281/zenodo.19885550 (v1, v2.0 superseded)
Drafting / forthcoming			
jSN	The j-invariant cell of degree-3 PCFs (standalone note, 4pp)	Polished; arXiv submission forthcoming	arXiv: TBA

ID	Title (working)	Venue / status	Zenodo concept / version DOI
D1	Cubic–quartic results paper	Drafting (post-PCF-2 v1.3)	–
D2-NOTE	Newton-polygon universality of $\xi_0 = d/\beta_d^{1/d}$	Drafting	–
D3	Channel-theory results paper	Planned (post-CT v1.3)	–
D7	AEAL methodology for verified experimental mathematics at scale	Drafting	–
MASTER-V0	SIARC Master Conjecture v0 announcement	Planned (post-T1 / Birkhoff–Trjitzinsky)	–
Carried over from v1 (status updated)			
#14	Four-tier classification of $\mathrm{PSL}_2(\mathbb{Z})$-uniformized PCFs	Submitted; under revision after two desk-rejects on citation hygiene	TBA (post-acceptance)
P08	Painlevé VI and trans-stratum PCFs (pointer paper)	Draft	–
G1	Extension of the four-tier hierarchy to $\Gamma_0(2)$	Draft	–
G2	Trans-stratum exponents for Hecke triangle groups	Outline	–
OP	Open-problems census for the SIARC program	Outline; superseded by §6 of this paper	–

Note on proofs. By design, no proof from any of the above papers is reproduced in this document. Where a claim in this document would require proof, it has been stated as a conjecture, an open problem, or a remark with a pointer to the responsible companion paper. Readers seeking proofs should consult the cited companion paper directly.

8 Closure Cascade and Bridge Provenance (v2.1 amendment)

This section is appended in v2.1 (May 2026). It records three milestone closures landed on the SIARC relay bridge (github.com/papanokechi/siarc-relay-bridge) since v2.0, with the canonical bridge SHA, the closure mechanism, and a pointer to the companion paper or bridge session that owns the underlying derivation. *No numerical claim in this section originates in this document*; every number is inherited from a companion paper or a previously deposited bridge session, traceable to a bridge `claims.jsonl` entry. The AEAL claim audit for v2.1 is recorded in `sessions/2026-05-09/T1-EXECUTOR-116-UMBRELLA-V21-M9-V0-DEPOSIT-133/claims.jsonl` on the bridge.

8.1 Milestone status table

Milestone	v2.0 status	v2.1 status (this amendment)	Bridge SHA(s)
M4 V0	PROVISIONAL (open mid-April 2026)	CLOSED via deg-a = 0 row mechanism (MEDIUM-HIGH; HIGH-pending Wasow §X.3 OCR + W21 LANE-1 ratification)	`5f9db69`
M6.CC R1	GATED on V_{quad} structural diagnosis	CLOSED via Q4 v2 cascade ($D_6 \to D_7$ degeneration boundary; HIGH conf.)	`10b5cf6` (packet 130), `8a22b11` (verdict 131)
Route F slot 115	PENDING numerical re-derivation	LANDED (dps = 300 re-derivation; s_1 fixed-point at $\alpha_1 = 0$)	`8ebd1eb`

8.2 Item 1 — M4 V0 closure (deg-$a = 0$ row mechanism)

The M4 stratum closure landed on the bridge as `M4-V0-CLOSURE-CASCADE-106` (bridge SHA `5f9db69`, 2026-05-08). The canonical synth-ratified closure statement (Claude Opus 4.7, T1-Synth, Claude.ai web; `ACCEPT_W_REVISIONS`) reads in part: *"M4 V0 CLOSED via the deg-$a = 0$ row mechanism (068 verdict UPGRADE_FULL_VIA_DEG_A_ZERO_ROW). The deg-$a = 0$ row supersedes the prior expected closure path (Costin–Borderline / Birkhoff–Trjitzinsky 1933 §1 anormal $q = 2$ fractional-rank resurgence) and provides the structural M4-stratum closure at zero acquisition cost. The closure runs at the algebraic-combinatorial level via the four-row Phase A WZ-balance enumeration extension and the V6 closed-form general formula $A_{\text{naive}} = 2d - d_a$, specialised to $d_a = 0$ to give $A_{\text{naive}} = 2d$ uniformly at general $d \geq 2$."* Confidence is recorded as *MEDIUM-HIGH; HIGH-pending* on post-W21-LANE-1 ratification + post-Wasow §X.3 OCR acquisition. Numerical residuals at the original 068 dispatch precision are $d = 3$ *cubic* $A_{\text{fit}} = 5.978 \pm 0.026$ (dps= 800, 50 families) and $d = 4$ *quartic* $A_{\text{fit}} = 7.954 \pm 0.0037$ (dps= 1200, 60 families), both within 1σ of the V6 prediction $A = 2d$ after the standard $1/\log N$ finite-window correction; these numbers are inherited from bridge session 068 and reproduced in `M4-V0-CLOSURE-CASCADE-106/m4_v0_closure_statement_canonical.md`.

8.3 Item 2 — M6.CC R1 closure (V_{quad} on the $D_6 \to D_7$ degeneration boundary, via the Q4 v2 cascade)

The M6.CC R1 closure landed on the bridge across two sessions: the Q4 round-2 substrate-direct derivation packet `T1-OPERATOR-Q4-ROUND2-DIRECT-DERIVATION-PACKET-130` (bridge SHA `10b5cf6`), and the synth verdict-absorption session `T1-SYNTH-Q4-V2-VERDICT-ABSORPTION-131` (bridge SHA `8a22b11`; verdict GO_ROUTE_F_FIXED_POINT_DISTINGUISHED HIGH conf.). The closure mechanism reframes the V_{quad} stratum as sitting on the $D_6 \to D_7$ degeneration boundary of Sakai's Painlevé III hierarchy: the V_{quad} image $(\eta_\infty, \eta_0, \theta_\infty, \theta_0) = (1/6, 0, 0, -1/2)$ violates the standing assumption $\eta_0 \neq 0$ of the $P'_{III}(D_6)$ generic open stratum, and the surviving Cremona symmetry reduces to $\widetilde{W}_a(A_1)$ on the $P'_{III}(D_7)$ sector. Substrate anchors: Okamoto [11] §1 (Hamiltonian conventions); Ohyama, Kawamuko, Sakai, Okamoto [12] §3.1 (the $D_6 \to D_7$ reduction map and the resulting D_7-type P_{III} Hamiltonian); Kajiwara, Noumi, Yamada [14] §8.5 (the $E_8^{(1)}$ discrete Painlevé Cremona-action ladder and the $\widetilde{W}_a(A_1)$ subgroup); Sakai [13] (rational-surfaces / affine-root-systems classification underpinning the $D_n^{(1)}$ surface labelling). This closure converges with the off-generic-stratum diagnosis of the 069r3 final synthesis (bridge SHA `ae5b7f7`, `T1-SYNTH-069R3-FINAL-VERDICT-ABSORPTION-124`): two independent T1-Synth threads land on the same structural feature (V_{quad} on the degenerate locus of $P_{III}(D_6)$) from different angles. The convergence is recorded as a positive forensic signal in `sessions/2026-05-09/T2-EXECUTOR-115-ROUTE-F-D7-FIXED-POINT-EXTRACTION-132/unexpected_finds.j`

8.4 Item 3 — Route F slot 115 numerical re-derivation

The Route F executor session `T2-EXECUTOR-115-ROUTE-F-D7-FIXED-POINT-EXTRACTION-132` (bridge SHA `8ebd1eb`, 2026-05-09) reproduces the synth Q4 v2.0 structural payload at agent-side numerical precision dps = 300. Pushing the V_{quad} image $(\eta_\infty, \eta_0, \theta_\infty, \theta_0) = (1/6, 0, 0, -1/2)$ through the Ohyama–Kawamuko–Sakai–Okamoto [12] §3.1 reduction map yields $(\alpha, \beta, \gamma, \delta) = (0, 0, 1/9, 0)$ with all residuals below 10^{-200} and the relation $\gamma\delta = 0$ (violating the generic $P'_{III}(D_6)$ standing assumption $\gamma\delta \neq 0$, the D_7-sector signature). On the $P'_{III}(D_7)$ sector the surviving Cremona action is $\widetilde{W}_a(A_1)$ with simple reflection $s_1 : \alpha_1 \mapsto -\alpha_1$; the V_{quad} image sits at $\alpha_1 = 0$, the s_1 fixed point, recovered as a closed-form mpmath identity. The companion paper PCF-1 [1] carries the manuscript pull-back: a new labelled remark `rem:vquad-d7-s1` in [1] §V_{quad} documents the $D_6 \to D_7$ sector descent and the s_1 fixed-point identity, with a cross-cascade convergence note pointing to the 069r3 off-generic-stratum diagnosis (bridge SHA `ae5b7f7`). The numerical re-derivation is reproducible from the script `vquad_p3d6_recovery.py::compute_d6_` and is logged as five AEAL claims in `sessions/2026-05-09/T2-EXECUTOR-115-ROUTE-F-D7-FIXED-POINT-EXT`

8.5 What v2.1 does *not* do

For clarity of scope: v2.1 does not modify §§1–7 in any way beyond the title, date, abstract addendum, and the par:v20-to-v21 paragraph; the cross-degree triple of §3, the Trans-Stratum Conjecture of §5, and the Open Problems of §6 are unchanged. The Route F closure does *not* resolve the v2.0 open problems (op:median-resurgence-extension, op:Pk-cells-structure, op:chi-Phi-compatibility); those remain open. The v2.1 amendment is a milestone-status update for the SIARC bridge, not a content revision of the program statement.

9 AI Disclosure

This v2.0 umbrella absorbs the post-March 2026 SIARC stack, which was produced via a *multi-agent fleet methodology*: parallel sessions for hypothesis-testing (the Fleet-A unification stack), audit fleets (Fleet-F), AEAL claim chains (per-session `claims.jsonl` files mirrored on the SIARC relay bridge github.com/papanokechi/siarc-relay-bridge), and rubber-duck critique passes on every release. All *mathematical content* in this document — definitions, conjectures, open problems, the structure of the program, the assignment of claims to companion papers, and the precise form of the cross-degree invariant triple of §3 — was determined by the author. The LLM tooling (GitHub Copilot powered by Anthropic Claude Opus 4.7, and standalone Anthropic Claude sessions) produced LaTeX scaffolding, numerical orchestration, build-log parsing, and copy-editing under direct human supervision; no part of the empirical census underlying Conjecture 11 or the Petersson-discriminant correlation of §3.3 was generated by a language model. Those numbers are the product of the deterministic computational pipelines described in T2B and PCF-2 v1.3, and are reproducible from the session deliverables on the SIARC bridge.

The AEAL claim schema — formalised in PCF-2 v1.3 ([2], the "AEAL schema (v1.3 canonical form)" paragraph; addresses Fleet-F audit finding F1-01) — is the canonical schema across the program from v1.3 forward, and is the schema used in the `sessions/2026-05-02/SIARC-UMBRELLA-V2-RELE` companion to this release. The author reviewed and takes full responsibility for every statement in this document.

Acknowledgments

The author thanks the maintainers of the Zenodo platform for providing a stable DOI infrastructure that makes a program-paper of this kind possible. v2.0 absorbs SIARC sessions PCF1-* (P12 / Compositio prep, January–April 2026), PCF2-* (R1.1, R1.2, R1.3, T2, V12-RELEASE, V13-RELEASE, May 2026), and the triple-fleet audit (Fleet-A, Fleet-F, standalone-note polish, May 2026). The author thanks the GitHub Copilot CLI multi-agent fleet for the synthesis, audit, and assembly support during the v1→v2.0 refactor.

References

[1] papanokechi, *Polynomial continued fractions at degree* $(d,1)$*: sharp WKB exponents and an arithmetic stratification* (SIARC companion paper PCF-1), Zenodo preprint, v1.3, 2026-05-01. Version DOI: 10.5281/zenodo.19937196; concept DOI: 10.5281/zenodo.19931635. Compositio Mathematica CM 10573 (in review).

[2] papanokechi, *PCF-2: Polynomial Continued Fractions at Degree Three — v1.3 (Cubic-Modular Framing)* (SIARC companion paper PCF-2), Zenodo preprint, v1.3, 2026-05-02. Version DOI: 10.5281/zenodo.19963298; concept DOI: 10.5281/zenodo.19936297.

[3] papanokechi, *Channel Theory:* V_{quad} *resurgence and the CC-pipeline* (SIARC companion paper CT), Zenodo preprint, v1.2, 2026-05-01. Version DOI: 10.5281/zenodo.19951331; concept DOI: 10.5281/zenodo.19941678.

[4] papanokechi, *Two arithmetic classes of degree-*$(2,1)$ *Trans-stratum continued fractions: a Birkhoff–Trjitzinsky / Gauss-continued-fraction dichotomy* (SIARC companion paper T2B), Zenodo preprint, v3.0, 2026-04-30. Version DOI: 10.5281/zenodo.19915689; concept DOI: 10.5281/zenodo.19783311.

[5] papanokechi, *The* j*-invariant cell of degree-*3 *polynomial continued fractions* (SIARC standalone note, 4pp), arXiv preprint, math.NT, forthcoming.

[6] papanokechi, *Four-tier classification of* $\mathrm{PSL}_2(\mathbb{Z})$*-uniformized polynomial continued fractions* (SIARC companion paper #14), submitted; DOI to be assigned upon acceptance.

[7] papanokechi, *Painlevé VI and trans-stratum polynomial continued fractions* (SIARC companion paper P08), draft; DOI to be assigned.

[8] G. D. Birkhoff, *Formal theory of irregular linear difference equations*, Acta Mathematica **54** (1930), 205–246.

[9] G. D. Birkhoff and W. J. Trjitzinsky, *Analytic theory of singular difference equations*, Acta Mathematica **60** (1933), 1–89.

[10] S. Chowla and A. Selberg, *On Epstein's zeta-function*, Journal für die reine und angewandte Mathematik **227** (1967), 86–110.

[11] K. Okamoto, *Studies on the Painlevé equations IV: third Painlevé equation* P_{III}. Funkcial. Ekvac. **30** (1987), 305–332.

[12] Y. Ohyama, H. Kawamuko, H. Sakai, K. Okamoto, *Studies on the Painlevé equations V, third Painlevé equations of types* D_7 *and* D_8. J. Math. Sci. Univ. Tokyo **13** (2006), 145–204.

[13] H. Sakai, *Rational surfaces associated with affine root systems and geometry of the Painlevé equations*. Comm. Math. Phys. **220** (2001), 165–229.

[14] K. Kajiwara, M. Noumi, Y. Yamada, *Geometric aspects of Painlevé equations*. J. Phys. A: Math. Theor. **50** (2017), 073001 (164 pp.).